

TABLE 1
BAN-Logic Symbols

Symbol	Description
$P \equiv X$	Principal P believes X .
$P \triangleleft X$	Principal P sees (receives) X .
$P \sim X$	Principal P once said X .
$P \Rightarrow X$	Principal P has jurisdiction over X .
$P \ni K$	Principal P possesses the key K .
$\#(X)$	Statement X is fresh.
$P \xleftrightarrow{K} Q$	K is a shared key between P and Q .
$\xrightarrow{K} P$	K is the public key belonging to P .
K^{-1}	Private key corresponding to public key K .

1 BAN LOGIC ANALYSIS

We performed a complete BAN logic analysis of the three components of the proposed framework: (1) Service subscription, (2) Re-encryption key distribution, and (3) End-to-end data-sharing. The full step-by-step derivation, applying the symbols and inference rules in Tables 1 and 2, is available in the public GitHub repository¹. Here, we summarize the goals that were formally validated by the analysis.

- 1) The vehicle believes it shares a valid session key with the Service Provider during the service subscription process.

$$V_i \models V_i \xleftrightarrow{K_s^+} SP \quad [Goal\ 1]$$

- 2) The SP believes it shares the same session key with the vehicle.

$$SP \models SP \xleftrightarrow{K_s^+} V_i \quad [Goal\ 2]$$

- 3) The vehicle believes it holds authentic and fresh group secret material.

$$V_i \models \langle x_i, w_{i,1}, w_{i,2}, w_{i,3} \rangle \quad [Goal\ 3]$$

- 4) The edge server believes that the SP authoritatively sends the re-encryption key and sequence number.

$$ES \models SP \models \langle rk, SN \rangle \quad [Goal\ 4]$$

- 5) The edge server accepts the re-encryption key as valid and fresh.

$$ES \models \langle rk, SN \rangle \quad [Goal\ 5]$$

- 6) The ES believes that the SP generates the ciphertext and is fresh.

$$ES \models SP \models \{H_2(m \parallel z^{x_2}), H_3(c_1), c_2\} \quad [Goal\ 6]$$

- 7) The ES believes that the SP indeed generated the ciphertext component c_1 .

$$ES \models SP \sim c_1 \quad [Goal\ 7]$$

- 8) The vehicle believes that the SP who sends the ciphertext and its is fresh.

$$V_i \models SP \models \{H_2(m \parallel z^{x_2}), H_3(c_1), c_2\} \quad [Goal\ 8]$$

- 9) The vehicle believes that the plaintext originated from the SP.

$$V_i \models SP \sim m \quad [Goal\ 9]$$

1. <https://github.com/Mahmoud-Abdelsatar/ESSDS>

TABLE 2
BAN-Logic Inference Rules

Rule	Description and Formal Rule
R1. Message-meaning (shared key)	$\frac{P \equiv P \xleftrightarrow{K} Q, P \triangleleft \{X\}_K}{P \equiv Q \mid \sim X}$
R2. Message-meaning (public key signature)	$\frac{P \equiv \xrightarrow{K} Q, P \triangleleft \{X\}_{K^{-1}}}{P \equiv Q \mid \sim X}$
R3. Nonce verification	$\frac{P \equiv \#(X), P \equiv Q \mid \sim X}{P \equiv Q \equiv X}$
R4. Jurisdiction rule	$\frac{P \equiv Q \Rightarrow X, P \equiv Q \equiv X}{P \equiv X}$
R5. Belief aggregation	$\frac{P \equiv X, P \equiv Y}{P \equiv (X, Y)}$
R6. Belief projection	$\frac{P \equiv (X, Y)}{P \equiv X}$
R7. Mutual belief projection	$\frac{P \equiv Q \equiv (X, Y)}{P \equiv Q \equiv X}$
R8. Said projection	$\frac{P \sim (X, Y)}{P \sim X}$
R9. Freshness propagation	1) $\frac{P \equiv \#(X)}{P \equiv \#(X, Y)}$ 2) $\frac{P \equiv \#(X), P \ni K}{P \equiv \#(\{X\}_K), P \equiv \#(\{X\}_{K^{-1}})}$
R10. Message decryption (unqualified key) [1]	$\frac{P \triangleleft X}{P \triangleleft X}$
R11. Possession rule [1]	$\frac{P \triangleleft X}{P \ni X}$
R12. Unqualified key-agreement rule [1]	$\frac{P \ni K_a^{-1}, P \ni \xrightarrow{K_b} Q}{P \ni K_s}$
R13. Qualified key-agreement rule [1]	$\frac{P \equiv K_a^{-1}, P \equiv \xrightarrow{K_b} Q}{P \equiv P \xleftrightarrow{K_s} Q}$
R14. Key confirmation rule [1]	$\frac{P \equiv P \xleftrightarrow{K_s} Q, P \equiv \#(X), P \triangleleft \{X\}_{K_s}}{P \equiv P \xleftrightarrow{K_s} Q}$
R15. Specialized nonce verification for certificates [1]	$\frac{P \equiv TA \mid \sim \langle Q, \xrightarrow{K_Q} Q, T_{exp} \rangle}{P \equiv TA \mid \sim \langle Q, \xrightarrow{K_Q} Q, T_{exp} \rangle}$
R16. Sight rule for public key encryption [2]	$\frac{P \equiv \xrightarrow{K_p} P, P \triangleleft \{X\}_{K_p}}{P \triangleleft X}$
R17. Sight projection	$\frac{P \triangleleft (X, Y)}{P \triangleleft X}$
R18. Sight rule for public key signature	$\frac{P \equiv \xrightarrow{K_q} Q, P \triangleleft \{X\}_{K_q^{-1}}}{P \triangleleft X}$
R19. Hash function rule [1]	$\frac{P \equiv Q \mid \sim H(X), P \triangleleft X}{P \equiv Q \mid \sim X}$

1.1 Assumption

- A1. $V_i \models \xrightarrow{K_{t_a}} TA$
- A2. $V_i \models K_a^{-1}$
- A3. $V_i \models \#(\xrightarrow{K_a} V_i)$
- A4. $V_i \models SP \mid \Rightarrow (\xrightarrow{K_b} SP)$
- A5. $SP \models \xrightarrow{K_{t_a}} TA$
- A6. $SP \models K_b^{-1}$

- A7. $SP \mid \equiv \#(\overset{K_b}{\mapsto} SP)$
- A8. $SP \mid \equiv Vi \mid \implies (\overset{K_a}{\mapsto} Vi)$
- A9. $ES \mid \equiv \overset{K_{sp}}{\mapsto} SP$
- A10. $ES \mid \equiv \#(T_s)$
- A11. $Vi \mid \equiv TA \mid \implies \langle SP, \overset{K_{sp}}{\mapsto} SP, T_{exp} \rangle$
- A12. $SP \mid \equiv TA \mid \implies \langle Vi, \overset{K_{vi}}{\mapsto} Vi, T_{exp} \rangle$
- A13. $Vi \mid \equiv \#(N)$
- A14. $Vi \mid \equiv SP \mid \implies \langle x_i, w_{i,1}, w_{i,2} \rangle$
- A15. $ES \mid \equiv SP \mid \implies \langle rk, SN \rangle$
- A16. $Vi \mid \equiv \#(SN)$
- A17. $ES \mid \equiv \#(SN)$
- A18. $Vi \mid \equiv \overset{K_{gpk}}{\mapsto} Vi$ where K_{gpk}^{-1} is calculated as $H(e(w_{i,1}, w_{i,2}))$
- A19. $Vi \ni K_a^{-1}$

1.2 Idealized Protocol

1.2.1 Service Subscription

Message 1. $Vi \rightarrow SP : \overset{K_a}{\mapsto} Vi$

Message 2. $SP \rightarrow Vi : (\overset{K_b}{\mapsto} SP, \{SP, \overset{K_{sp}}{\mapsto} SP, T_{exp}\}_{K_{ta}^{-1}}, \{\{\overset{K_a}{\mapsto} Vi, \overset{K_b}{\mapsto} SP\}_{K_{sp}^{-1}}\}_{K_s})$

Message 3. $Vi \rightarrow SP : (\{Vi, \overset{K_{vi}}{\mapsto} Vi, T_{exp}\}_{K_{ta}^{-1}}, \{\{\overset{K_a}{\mapsto} Vi, \overset{K_b}{\mapsto} SP, N\}_{K_{vi}^{-1}}\}_{K_s})$

Message 4. $SP \rightarrow Vi : \{x_i, w_{i,1}, w_{i,2}, N\}_{K_s}$

It is worth mentioning that $\overset{K_a}{\mapsto} Vi$ and $\overset{K_b}{\mapsto} SP$ are ephemeral keys while the $\overset{K_{vi}}{\mapsto} Vi$ and $\overset{K_{sp}}{\mapsto} SP$ are long-term signing keys.

1.2.2 Re-encryption Key Distribution

Message 1. $SP \rightarrow ES : \{rk, SN\}_{K_{sp}^{-1}}$

1.2.3 Data Sharing

Message 1. $SP \rightarrow ES : c_1, \{H_2(m \parallel z^{x_2}), H_3(c_1), c_2, SN\}_{K_{sp}^{-1}}$

Re-encrypted Message 1. $ES \rightarrow Vi : c_1', c_2, \{H_2(m \parallel z^{x_2}), H_3(c_1), c_2, SN\}_{K_{sp}^{-1}}$

1.3 Analysis

1.3.1 Service Subscription

Step 1. From Message2, we have:

$$Vi \triangleleft \overset{K_b}{\mapsto} SP \quad (1)$$

Applying the possession rule R11 using (1), we have:

$$Vi \ni \overset{K_b}{\mapsto} SP \quad (2)$$

Step 2. Using A19 and (2), Vi computes $K_s = K_b^a$. By applying the unqualified key rule R12, we have:

$$Vi \ni K_s \quad (3)$$

Step 3. From message 2, we have:

$$Vi \triangleleft \{SP, \overset{K_{sp}}{\mapsto} SP, T_{exp}\}_{K_{ta}^{-1}} \quad (4)$$

Applying the message-meaning rule R2 on (4) and A1, we get:

$$Vi \mid \equiv TA \mid \sim \langle SP, \overset{K_{sp}}{\mapsto} SP, T_{exp} \rangle \quad (5)$$

Step 4. Then, applying R15 on (5) results in:

$$Vi \mid \equiv TA \mid \equiv \langle SP, \overset{K_{sp}}{\mapsto} SP, T_{exp} \rangle \quad (6)$$

Step 5. By applying jurisdiction Rule, R4, using A11 and (6), we get:

$$Vi \mid \equiv \langle SP, \overset{K_{sp}}{\mapsto} SP, T_{exp} \rangle \quad (7)$$

Step 6. By applying the belief projection, R6, on (7), we get:

$$Vi \mid \equiv \overset{K_{sp}}{\mapsto} SP \quad (8)$$

Step 7. From message 2, we have:

$$Vi \triangleleft \{\{\overset{K_a}{\mapsto} Vi, \overset{K_b}{\mapsto} SP\}_{K_{sp}^{-1}}\}_{K_s} \quad (9)$$

By applying R10 to (3) and (9), we get:

$$Vi \triangleleft \{\{\overset{K_a}{\mapsto} Vi, \overset{K_b}{\mapsto} SP\}_{K_{sp}^{-1}}\}_{K_s} \quad (10)$$

Step 8. Applying R2 to (8) and (10) results in:

$$Vi \mid \equiv SP \mid \sim \{\overset{K_a}{\mapsto} Vi, \overset{K_b}{\mapsto} SP\} \quad (11)$$

Step 9. Applying freshness propagation rule, R9.1, on A3 and (11), we get:

$$Vi \mid \equiv \#(\{\overset{K_a}{\mapsto} Vi, \overset{K_b}{\mapsto} SP\}) \quad (12)$$

Step 10. Applying the nonce-verification rule R3 on (11) and (12), we get:

$$Vi \mid \equiv SP \mid \equiv \{\{\overset{K_a}{\mapsto} Vi, \overset{K_b}{\mapsto} SP\}\} \quad (13)$$

Step 11. Then, applying the mutual belief projection rule R7 on (13), we get:

$$Vi \mid \equiv SP \mid \equiv \overset{K_b}{\mapsto} SP \quad (14)$$

Step 12. Applying Jurisdiction rule R4 on A4 and (14), we get:

$$Vi \mid \equiv \overset{K_b}{\mapsto} SP \quad (15)$$

Step 13. Applying Qualified key rule, R13, on A2 and (15), we get:

$$Vi \mid \equiv Vi \overset{K_s}{\mapsto} SP \quad (16)$$

Step 14. Applying R18 to A1 and (4), we get:

$$Vi \triangleleft SP, \overset{K_{sp}}{\mapsto} SP, T_{exp} \quad (17)$$

Then applying R17 to (17), we get:

$$Vi \triangleleft \overset{K_{sp}}{\mapsto} SP \quad (18)$$

and applying the possession rule, R11, using (18), results in:

$$Vi \ni \overset{K_{sp}}{\mapsto} SP \quad (19)$$

Therefore, by applying the freshness propagation R9.2 to (12) and (19), we get:

$$Vi \equiv \#(\{ \xrightarrow{K_a} Vi, \xrightarrow{K_b} SP \}_{K_{sp}^{-1}}) \quad (20)$$

Step 15. By applying the confirmed key rule, R14, on (9),(16), and (20), we get:

$$Vi \equiv Vi \xleftrightarrow{K_{vi}^+} SP \quad G1 \quad (21)$$

Step 16. From message 3, we have:

$$SP \triangleleft \{ Vi, \xrightarrow{K_{vi}} Vi, T_{exp} \}_{K_{ta}^{-1}} \quad (22)$$

Repeating steps 3-6 by applying R2,R15, R4, R6,respectively, starting by A5 and (22), results in:

$$SP \equiv \xrightarrow{K_{vi}} Vi \quad (23)$$

Step 17. From message 1, we have:

$$SP \triangleleft \xrightarrow{K_a} Vi \quad (24)$$

By applying the possession rule, R11, we get:

$$SP \ni \xrightarrow{K_a} Vi \quad (25)$$

Step 18. From message 3, we have:

$$SP \triangleleft \{ \{ \{ \xrightarrow{K_a} Vi, \xrightarrow{K_b} SP, N \}_{K_{vi}^{-1}} \}_{K_s} \quad (26)$$

Repeating steps 7-15 by applying R10, R2, R9.1, R3, R7, R4, R13, R18, R17, R11, R9.2, and R14, respectively, starting from (26), results in:

$$SP \equiv SP \xleftrightarrow{K_{vi}^+} Vi \quad G2 \quad (27)$$

Step 19. From message 4, we have:

$$Vi \triangleleft \{ x_i, w_{i,1}, w_{i,2}, N \}_{K_s} \quad (28)$$

By applying R1 using (21) and (28), we get:

$$Vi \equiv SP \mid \sim \{ x_i, w_{i,1}, w_{i,2}, N \} \quad (29)$$

Step 20. Applying R9.1, using A13, results in:

$$Vi \equiv \#(\{ x_i, w_{i,1}, w_{i,2}, N \}) \quad (30)$$

Step 21. By applying R3 to (29) and (30), we get:

$$Vi \equiv SP \mid \equiv \{ x_i, w_{i,1}, w_{i,2}, N \} \quad (31)$$

Then, applying the mutual belief projection rule R7 to (31), we get:

$$Vi \equiv SP \mid \equiv \langle x_i, w_{i,1}, w_{i,2} \rangle \quad (32)$$

Step 22. Applying R4 using A14 and (32), we get:

$$Vi \equiv \langle x_i, w_{i,1}, w_{i,2} \rangle \quad G3 \quad (33)$$

1.3.2 Re-encryption Key Distribution

Step 1. From message 1, we have

$$ES \triangleleft \{ rk, SN \}_{K_{sp}^{-1}} \quad (34)$$

Step 2. By applying R2 using A9 and (34), we get:

$$ES \mid \equiv SP \mid \sim \{ rk, SN \} \quad (35)$$

Step 3. Applying R9.1 on A17, results in:

$$ES \mid \equiv \#(rk, SN) \quad (36)$$

Then, applying R3 to (35) and (36) results in:

$$ES \mid \equiv SP \mid \equiv \langle rk, SN \rangle \quad G4 \quad (37)$$

Step 4. By applying R4 using A15 and (37), we get:

$$ES \mid \equiv \langle rk, SN \rangle \quad G5 \quad (38)$$

1.3.3 Data Sharing

Step 1. From message 1, we have:

$$ES \triangleleft c1 \quad (39)$$

Step 2. From message 1, we have:

$$ES \triangleleft \{ H_2(m \parallel z^{x_2}), H_3(c_1), c_2, SN \}_{K_{sp}^{-1}} \quad (40)$$

Step 3. By applying R2 using A9 and (40), we get:

$$ES \mid \equiv SP \mid \sim \{ H_2(m \parallel z^{x_2}), H_3(c_1), c_2, SN \} \quad (41)$$

Step 4. By applying R9.1 on A17, we get:

$$ES \mid \equiv \#(\{ H_2(m \parallel z^{x_2}), H_3(c_1), c_2, SN \}) \quad (42)$$

Step 5. Applying R3, using (41) and (42), results in:

$$ES \mid \equiv SP \mid \equiv \{ H_2(m \parallel z^{x_2}), H_3(c_1), c_2, SN \} \quad (43)$$

Then, applying R7 to (43) results in:

$$ES \mid \equiv SP \mid \equiv \{ H_2(m \parallel z^{x_2}), H_3(c_1), c_2 \} \quad G6 \quad (44)$$

Step 6. Applying R8 to (41), we get:

$$ES \mid \equiv SP \mid \sim H_3(c_1) \quad (45)$$

Applying R19 to (39) and (45), we get:

$$ES \mid \equiv SP \mid \sim c1 \quad G7 \quad (46)$$

By applying **Step 7.** From re-encrypted message 1, we have:

$$Vi \triangleleft c1' \quad \text{where } c1' = \{ z^{x_1} \}_{K_{gpk}} \quad (47)$$

Applying R16 using A18 and (47) results in:

$$Vi \triangleleft z^{x_1} \quad (48)$$

And applying R11 to (48) results in:

$$Vi \ni z^{x_1} \quad (49)$$

Step 8. From re-encrypted message 1, we have:

$$Vi \triangleleft c2 \quad \text{where } c2 = \{ m \parallel z^{x_2} \}_{z_{x_1}} \quad (50)$$

Applying R10 using (49) and (50) results in:

$$Vi \triangleleft m \parallel z^{x_2} \quad (51)$$

Step 9. Applying the sight projection rule R17 on (51) results in:

$$Vi \triangleleft m \quad (52)$$

Step 10. From re-encrypted message 1, we have:

$$Vi \triangleleft \{H_2(m \parallel z^{x_2}), H_3(c_1), c_2, SN\}_{K_{sp}^{-1}} \quad (53)$$

Applying R2 using (8) and (53) results in:

$$Vi \equiv SP \mid \sim \{H_2(m \parallel z^{x_2}), H_3(c_1), c_2, SN\} \quad (54)$$

Step 11. Applying R9.1 on A16, we get:

$$Vi \equiv \#(\{H_2(m \parallel z^{x_2}), H_3(c_1), c_2, SN\}) \quad (55)$$

Step 12. Applying R3 to (54) and (55) results in:

$$Vi \equiv SP \mid \equiv \{H_2(m \parallel z^{x_2}), H_3(c_1), c_2, SN\} \quad (56)$$

Then applying R7 to (56) results in:

$$Vi \equiv SP \mid \equiv \{H_2(m \parallel z^{x_2}), H_3(c_1), c_2\} \quad G8 \quad (57)$$

Step 13. Again, applying R8 on (54), we get:

$$Vi \equiv SP \mid \sim H_2(m \parallel z^{x_2}) \quad (58)$$

Then, applying R19 to (52) and (58), we get

$$Vi \equiv SP \mid \sim m \quad G9 \quad (59)$$

REFERENCES

- [1] P. van Oorschot, "Extending cryptographic logics of belief to key agreement protocols," in *Proceedings of the 1st ACM Conference on Computer and Communications Security*, pp. 232–243, 1993.
- [2] M. Burrows, M. Abadi, and R. Needham, "A logic of authentication," *ACM Transactions on Computer Systems (TOCS)*, vol. 8, no. 1, pp. 18–36, 1990.